

Chasing the Surge

Predicting NYC Taxi Demand for Fleet Positioning

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Github repo with commit

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1 Introduction

New York City’s taxi fleet completed over 110 million trips between January 2024 and May 2026 [1], yet drivers routinely miss fares not from a lack of demand, but from waiting in the wrong zone when that demand appears. This is hard to predict, since weather, events, and public holidays all shift demand away from a zone’s routine history, motivating our research question: to what extent can historical taxi demand, weather, major events, public holidays, and temporal patterns predict where and when demand will be unusually high, and how can those predictions support fleet positioning decisions?

In this report, we measure how much weather, events, and public holidays improve demand predictions over a historical baseline, and use this to guide where fleets should position drivers before a shift, rather than reacting after demand shifts.

1.1 Target Audience and Datasets

This report is written for fleet operators and dispatchers, since they are the ones positioned to act on a citywide prediction by directing many drivers at once, not an individual driver who can only react to their own immediate surroundings. Every recommendation is built around one dispatch question: which zone-hours should the fleet prioritise this shift? Throughout the report, demand is measured at a zone-hour grain, one taxi zone in a given hour, as the number of trips picked up there.

The primary source is TLC trip records for Yellow and Green taxis, covering January 2024 to May 2026. Both taxi types are included, not Yellow alone, since Green serves the outer-borough and upper-Manhattan zones Yellow has historically underserved. Restricting to Yellow would exclude exactly the areas where fleet positioning has the most room to add value. This 29-month span covers every season at least twice, letting weather and holiday effects be observed under varied conditions, and leaves a forward-looking test period in 2026 rather than a random split, beyond the 6-month minimum.

Dataset	Rows (raw)	Features	Source
TLC Yellow + Green trips	110,354,209	19–21	TLC Trip Record Data [1]
Hourly weather (LGA)	21,122	11	Iowa Env. Mesonet ASOS [2]
NYC permitted events	629,145	13	NYC Open Data [3]
US federal holidays	34 (2024–2026)	2	US OPM [4]

Table 1: Datasets used in this study.

1.2 Features of Interest

The prediction target is the number of trips picked up per zone-hour, a continuous count, not a categorical label. Predictors map directly onto the research question: a lagged historical baseline, weather, event activity, holiday flags, and time itself, expressed as hour, day of week, and month. The historical baseline is computed only from a zone-hour’s own prior occurrences, never from its own or a future observation, so no predictor can leak outcome information.

Unusually high demand is defined through the Demand to Disruption Index, a continuous ratio of actual demand to a zone-hour’s own historical baseline:

$$\text{DTDI} = \frac{\text{demand}}{\text{baseline}} \quad (1)$$

Because DTDI is continuous rather than a fixed category, Section 4 treats demand prediction as a regression problem rather than a classification one.

1.3 Methodology Overview

Raw TLC trip records for Yellow and Green taxis are cleaned and joined with weather, event, and holiday data at a zone-hour grain, alongside a historical baseline computed for each zone-hour slot. The joined table is explored for temporal and geospatial demand patterns, then used to train and compare several regression models for predicting the number of trips picked up at zone-hour grain, headlined by an interpretable Linear Regression against a more flexible Gradient Boosting model. Model outputs are evaluated on both raw accuracy and how well each model ranks the busiest zone-hours, and the results are translated into concrete fleet positioning recommendations.

2 Preprocessing

2.1 TLC Taxi Data

Each row-removal rule below is grounded in the Yellow and Green Taxi Trip Records Data Dictionaries; Table 2 gives the resulting shape at each stage.

- **Missing or invalid location:** null pickup time/zone, or a zone ID outside the valid range of 265 zones, since location is essential and cannot be inferred.
- **Zero or negative distance** (2,822,289 rows, 2.56%): physically impossible for a real trip.
- **Negative fare** (3,386,984 rows, 3.07%): violates TLC fare rules.
- **Invalid duration** (765,835 rows, 0.69%): zero minutes or below, or over 24 hours which could be the result of a meter or logging error.
- **Duplicates and out-of-window rows:** duplicates and any row outside Jan 2024 to May 2026

2.2 External Data

- **Weather:** joins directly by pickup hour (99.76% coverage; final day falls just outside the source range).
- **Events:** expanded from each start-to-end span into every hourly zone-hour cell covered (26.95% of zone-hours have ≥ 1 event).
- **Holidays:** join by calendar date (2.95% of zone-hours).

Table 2 summarises the resulting shape of the TLC taxi pipeline at each stage.

Stage	Yellow rows	Green rows	Combined
Raw (Landing = Raw, no loss)	108,891,604	1,462,605	110,354,209
Clean (after filtering above)	101,986,947	1,392,075	103,379,022
Processed (zone-hour, incl. zero-demand)	5,567,184	5,567,184	11,134,368

Table 2: Dataset shape at each pipeline stage. Clean removes 6.32% of rows; zero-demand rows then dominate Processed, especially for Green (92.6% vs. Yellow’s 46.4%).

2.3 Aggregation and Historical Baseline

2.3.1 Aggregation

Distance and duration are flagged, not removed, as extreme-value outliers, using a 99th-percentile threshold computed separately for each taxi type and hour, since an unusually long trip is still real. Trips are aggregated to a pickup count per (taxi type, zone, hour). Every zero-trip zone-hour is added explicitly as a real “zero demand” observation, so the grid covers all 263 geographic zones (of 265 valid TLC location IDs, 2 are non-geographic placeholders) for the full study period.

2.3.2 Historical baseline

The baseline for each (taxi type, zone, day-of-week, hour) slot is the mean pickup count over the trailing 8 weeks, computed causally so it never uses its own or a future observation, keeping it leakage-safe and responsive to recent trends. A hierarchical fallback (coarser zone/hour grain, then a static borough average) covers slots without 8 weeks of history yet.

This rolling design replaced an earlier static (single 2024–2025 average) baseline, after that version’s own deviation map showed JFK and LaGuardia running persistently below their historical average despite very high raw demand (Figure 1, left), since a compressed multi-year average failed to track a real demand trend at the airports, making any calmer recent period look artificially “below normal” by comparison. The rolling baseline resolves this. JFK’s deviation moves from -3.95 to -0.07 pickups per hour and LaGuardia’s from -3.10 to $+0.86$ (Figure 1, right), both now close to neutral.

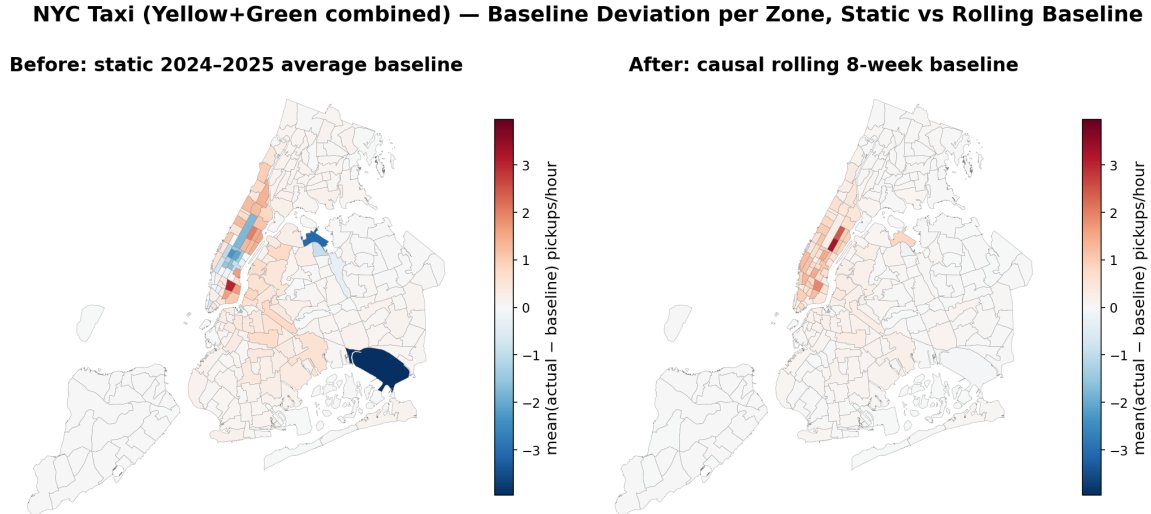


Figure 1: NYC taxi (Yellow+Green) baseline deviation per zone, static (before) vs. rolling (after). JFK/LaGuardia’s anomaly (dark blue, left) resolves under the rolling baseline.

3 Analysis and Geospatial Visualisation

3.1 Distribution and Spatial Concentration

Demand per zone-hour is heavily right-skewed: the median zone-hour sees a single pickup, 45.9% are exactly zero, and the mean is pulled to 18.5 by a long tail, with the busiest 1% exceeding 275 pickups up to a citywide maximum of 1,085 (Figure 2a). Yellow dominates this total (Table 2), so the combined distribution is Yellow-shaped, Green only thinning the tail. No outlier trimming is applied, since unusually high demand is the signal predicted here, not noise. Demand is also spatially concentrated (Figure 2b): Manhattan and the JFK/LaGuardia zones average roughly 44 times outer-borough demand. This reflects Manhattan’s density and parking scarcity, and JFK/LaGuardia’s near-total reliance on airport transfers, motivating zone as a model input.

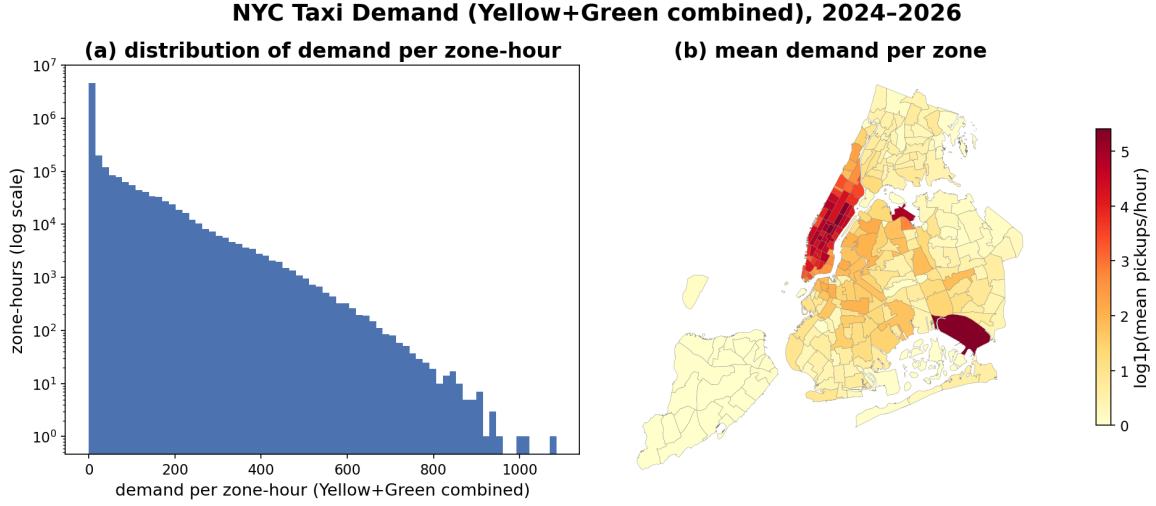


Figure 2: NYC taxi demand (Yellow+Green combined), 2024–2026. (a) Distribution of demand per zone-hour (log scale). (b) Mean demand per zone (log scale).

3.2 Temporal Patterns

Temporal patterns are grain-dependent (Figure 3), driven by weekday commuting and weekend nightlife. Day-of-week grain shows a 28% swing: Monday trails every other day, from its low of 15.8 mean pickups per hour to Thursday’s high of 20.1. Hour-of-day grain adds a midnight pattern: Saturday and Sunday run higher than any weekday at the same hour, before all seven days converge to a predawn trough. This reverses at the evening commute: weekdays peak at 31.5 mean pickups per hour (6 pm), while the weekend peak is flatter and slightly lower, around 28.1, at that same hour.

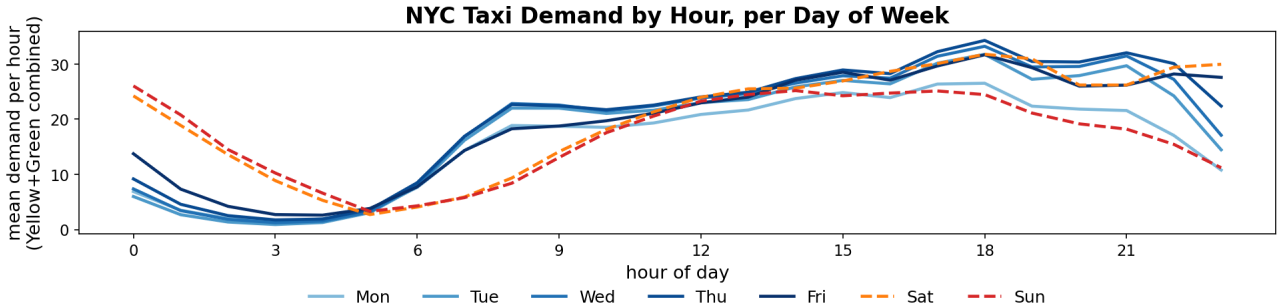


Figure 3: Mean NYC taxi demand (Yellow+Green combined) by hour of day, one line per day of week, weekdays shaded light to dark blue, weekend shown as dashed orange and red lines.

3.3 Weather and Event Relationships

Weather and event context show two different shapes (Figure 4). Demand versus temperature is flat from roughly -15 to 20°C , around 8 to 9.3 mean pickups per hour, then rises sharply above 25°C , nearly doubling by 35°C . This is consistent with extreme heat pushing more people into a cab, though it is a seasonal correlation, not proof of a direct causal effect: summer months also bring higher tourism and activity too. This heat effect is also distinct from severe weather: storms and snow do not show the same demand increase. Demand versus event count tells a sharper story. Any concurrent permitted event (protests, filming, block parties, and similar) roughly doubles mean demand, from 7.2 pickups per hour with none to 12 to 16 with one or more, consistent with events concentrating foot traffic and reducing walking or transit as viable alternatives in the immediate area.

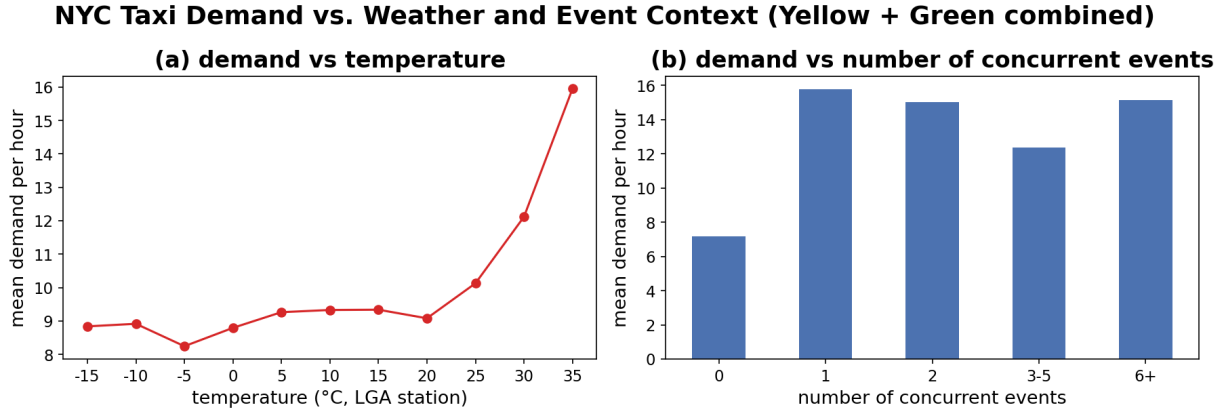


Figure 4: NYC taxi demand (Yellow+Green combined). (a) Demand vs. temperature. (b) Demand vs. concurrent event count.

4 Modelling

4.1 Prediction Target and Setup

We predict $y = \log(1 + \text{pickup count})$, which stabilises the right-skewed target and guarantees a non-negative back-transformed prediction. This continuous target means every model here is a regression, not a classifier. Predictors span three attribute types, continuous (the rolling baseline, event count, and raw weather numerics), categorical (zone, hour of day, day of week, month, weather severity, taxi type), and boolean (event presence, holiday flag). Section 1.2 lists the full predictor set, including the baseline’s leakage rule, which holds for every model here.

Taxi type is still included as a per-row predictor, since Yellow and Green differ structurally in scale, even though all reporting elsewhere treats demand as one combined total. Every model trains only on 2024–2025 and evaluates on 2026 alone, strictly future data, never a random or cross-validated split, consistent with the project’s prediction requirement.

4.2 Models

Four models were fit.

Linear Regression is an interpretable benchmark: it predicts the response as a weighted sum of the predictors, with each coefficient directly readable as that predictor’s effect on demand.

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} \quad (2)$$

Gradient Boosting (histogram-based, [5]) is an ensemble of shallow trees fit sequentially to residuals. It is set as a counterpart to Linear Regression, chosen for its ability to capture non-linear interactions, such as weather effects that only matter at certain hours, without manual feature engineering.

Ridge and LASSO Regression [6] (L2 and L1 penalties) were also fit as secondary comparisons. LASSO’s regularisation strength, α , was tuned on a held-out validation slice from the final two months of training, never the 2026 test set; the tuned value, $\alpha = 0.001$, is examined in Section 5 for which features it keeps.

4.3 Results

Table 3 reports each model’s accuracy on the 2026 test set. Gradient Boosting leads on all three metrics, mean absolute error (MAE), root mean squared error (RMSE), and R^2 , though its edge over Linear Regression and Ridge on RMSE and R^2 is marginal; LASSO trails on all three.

Model	MAE	RMSE	R^2
Gradient Boosting	2.20	10.09	0.934
Linear Regression	2.30	10.16	0.933
Ridge Regression	2.30	10.16	0.933
LASSO Regression	2.34	10.39	0.929

Table 3: Regression metrics on the 2026 test set

5 Discussion

5.1 Accuracy vs. Ranking Quality

Table 3 shows Gradient Boosting leads narrowly on every raw accuracy metric, but the margin over Linear Regression and Ridge is too small to be read as decisive. The gap widens once the metric matches the operational task: fleet operators care which zone-hours to prioritise, not the exact count, so ranking quality matters more than point accuracy. Measuring each model’s top- k overlap with the actual top- k (Table 4), Linear Regression and LASSO both rank *better* than Gradient Boosting at 3 of 4 k ’s, despite its marginal accuracy edge; only at $k = 50$ does Gradient Boosting edge ahead of Linear Regression and match LASSO. The two models are used for different jobs, not one crowned “best”.

k	Random	Linear/Ridge	GB	LASSO
50	0.003%	38%	40%	40%
200	0.011%	55%	42%	56%
1,000	0.05%	60%	50%	59%
5,000	0.27%	75%	73%	76%

Table 4: Top- k overlap: % of the true top- k busiest zone-hours each model’s ranking recovers.

5.2 Feature Importance: Two Perspectives

Table 5 offers two views on the same question: once a zone-hour’s baseline is known, what else matters? Gradient Boosting ranks hour of day as the top secondary factor, followed by day of week, holiday flag, month, and zone, with weather and events contributing little. LASSO disagrees on hour and day, eliminating both entirely, but this is not a contradiction. The baseline is already computed *per* (zone, day of week, hour), so any further *additive linear* value from hour or day is genuinely near zero, which is exactly what LASSO detects, while Gradient Boosting’s advantage comes from *nonlinear* interactions, such as hour combined with zone or weather, that no linear coefficient can show. LASSO does retain month and the holiday flag, the two temporal signals the baseline does not already encode, confirming they were worth engineering separately.

Feature group	GB permutation importance	LASSO status ($\alpha = 0.001$)
Baseline	1.817 (dominant, $392\times$)	Retained (coef. 1.21)
Hour of day	0.0046 (top factor)	Eliminated (24 levels)
Day of week	0.0017	Eliminated (7 levels)
Holiday flag	0.0016	Retained
Month	0.0014	Retained (7/12 levels)
Zone / Borough	0.0008	Retained (4/6 boroughs)
Taxi type	0.0003	Retained (Both)
Weather (numeric)	0.00025	Retained (2/7 levels)
Events	0.00005	Eliminated (2 levels)
Weather severity	0.000026	Eliminated (3 levels)

Table 5: Gradient Boosting permutation importance, computed on a 20,000-row subsample of the test set since permutation importance re-predicts the full sample once per feature per repeat, computationally prohibitive at the full 1.89M-row scale, vs. LASSO feature selection, by feature group. GB sees full zone identity (263 zones); LASSO sees a coarser 6-category borough grouping.

5.3 Prediction Error Analysis

Table 6 examines model error at the zone-hour level through DTDI, the ratio of actual demand to baseline. Predicted DTDI is far smoother than actual DTDI for every model, but by very different amounts: Gradient Boosting’s predicted DTDI has roughly $5\times$ smaller standard deviation than actual DTDI, while Linear Regression and Ridge are compressed over $20\times$ and LASSO over $28\times$, all regressing toward the baseline more than real demand does, a structural smoothness property of regression models rather than a fixable bug. Because of this compression, a 4-way DTDI classification (below normal, normal, elevated, very elevated) was tested and found structurally unreliable at recovering the upper categories, so it plays no evidentiary role in this report.

This same compression explains a second finding: none of the four models reliably capture how demand actually shifts under severe weather. Severe-weather zone-hours see MAE rise by roughly $1.6\times$ across every model (Table 6), and actual demand runs measurably *below* baseline in severe weather (mean DTDI 0.898, vs. 1.000 in normal conditions) rather than spiking, the opposite of what a naive “bad weather means busier streets” assumption would predict. Yet predicted DTDI barely moves, and not always in the right direction: Gradient Boosting, the least-compressed model, recovers about 14% of the true drop, while Linear Regression, Ridge, and LASSO’s predicted DTDI actually rises slightly instead of falling. These are exactly the models with the most compressed predictions in the first place. Section 6’s ranking-based evidence does not share this weakness, since it only requires correctly *ordering* zone-hours, not matching the full spread of the actual distribution.

Model	DTDI compression	Severe MAE ratio	Drop captured
Gradient Boosting	$4.89\times$	$1.60\times$	$\sim 14\%$
Linear Regression	$20.41\times$	$1.61\times$	-2%
Ridge Regression	$20.41\times$	$1.61\times$	-2%
LASSO Regression	$28.25\times$	$1.59\times$	-5%

Table 6: Model reliability under severe weather. DTDI compression: actual DTDI’s standard deviation divided by each model’s predicted DTDI standard deviation (larger = more compressed). Severe MAE ratio: severe-weather MAE divided by normal-weather MAE. Drop captured: how much of the true 0.103 mean-DTDI drop (1.000 normal $\rightarrow$ 0.898 severe) each model’s own predictions reproduce; a negative value means predicted DTDI moved the wrong way.

6 Recommendations

6.1 Recommendation 1 – rank zone-hours, don’t just predict a number

Fleet operators should generate a daily ranked list of predicted busiest zone-hours using Linear Regression or LASSO, not Gradient Boosting, and prioritise the top 20 to 50 zone-hours each shift. This carries no meaningful accuracy trade-off: Table 3 shows Linear Regression’s RMSE and R^2 trail Gradient Boosting only marginally, the same small gap discussed in Section 5.1, and Table 4 shows it ranks better at 3 of the 4 k ’s tested (Gradient Boosting only narrowly edges ahead at $k = 50$), so it remains the clear choice for this decision.

Table 4 shows why. Linear Regression recovers 38% of the true top-50 busiest zone-hours and 60% of the top-1000, against a random baseline of 0.003% and 0.05%, over $1,000\times$ better. LASSO leads/ties at 3 of 4 k ’s despite the worst raw accuracy of any regression model. On the busiest test-period day, 25 April 2026, a top-20 list correctly identified 13 of 20 actual busiest zone-hours, concentrated in the same Manhattan core (East/West Village, Upper East Side) that Section 3.1 already found running roughly 44 times outer-borough demand. This is a specific, practical workflow: a named model, a shift-usable list size, and a measured hit rate an operator can act on directly.

6.2 Recommendation 2 – severe weather warns but does not spike demand

Two actions follow from how models behave under severe weather. Firstly, dispatch should not surge-position drivers expecting a demand spike. Secondly, operators should treat whichever model’s predicted count as unreliable by roughly $1.6\times$ its normal error margin, leaning toward the lower end of that range since actual demand runs below the prediction, not above. Both follow from the same finding: severe weather does not spike demand, and none of the four models reliably detect the shift that does happen, the opposite of what an operator naively expecting a weather-driven surge assumes.

Mean DTDI falls to 0.898 in severe weather, below the 1.000 normal-conditions level (Table 6). Every model’s predictions barely move in response, and three of the four move the wrong way entirely: Gradient Boosting recovers about 14% of this drop, while Linear Regression, Ridge, and LASSO’s predicted DTDI rises slightly instead of falling, since their predictions are 20 to 28 times more compressed than actual demand variation. Every model’s own accuracy also degrades by roughly $1.6\times$ MAE under these conditions (Table 6), confirming this is a real, model-confirmed limitation, not an artefact of one severe-weather day. Surge-positioning on top of an already-optimistic number would only misallocate drivers further, away from zone-hours that are actually busy.

7 Conclusion

Four models predicted NYC taxi demand at a zone-hour grain from historical, weather, event, and holiday data. Gradient Boosting leads narrowly on every raw accuracy metric, but Linear Regression’s and LASSO’s superior ranking quality is what actually supports the recommended rank-based positioning, since “most accurate” and “most operationally useful” are not the same model here. A severe-weather caution, grounded in every model’s failure to capture the actual demand shift, addresses the second recommendation. Limitations here span both scope and method: no fare data was used, so recommendations are operational rather than financial, and the DTDI-classification signal proved structurally unreliable and was not used as evidence. Future work could add a year-over-year trend feature, incorporate built-environment characteristics (e.g. hotel and office density) to distinguish zones with similar demand history but different drivers, or calibrate per-zone thresholds on a model’s own predicted distribution rather than the actual one.

References

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- [5] Jerome H. Friedman. “Greedy function approximation: A gradient boosting machine”. In: *Annals of Statistics* 29.5 (2001), pp. 1189–1232.
- [6] F. Pedregosa et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.